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Original practice paper

WAEC Biology: Objective Practice 1

Fifteen original objective questions in the style of the WASSCE, across cells, nutrition, ecology, genetics and body systems.

Exam	Subject	Questions	Time
WAEC WASSCE	**Biology**	**15**	**15 minutes**

Instructions

- Answer every question.
- Each question carries one mark.
- Explanations follow the answer grid.

1. Which organelle releases energy from food in a cell?

A Nucleus
B Vacuole
C Ribosome
D Mitochondrion

2. Which structure is found in plant cells but NOT in animal cells?

A Cell membrane
B Cytoplasm
C Cell wall
D Nucleus

3. The green pigment that traps light energy in leaves is

A keratin
B melanin
C chlorophyll
D haemoglobin

4. Which gas is given off during photosynthesis?

A Hydrogen
B Nitrogen
C Oxygen
D Carbon dioxide

5. During vigorous exercise, anaerobic respiration in human muscle produces

A ethanol
B glucose
C oxygen
D lactic acid

6. An organism that makes its own food from simple substances is called a

A producer
B consumer
C decomposer
D parasite

7. In the food chain grass → grasshopper → lizard → hawk, the lizard is a

A tertiary consumer
B secondary consumer
C producer
D primary consumer

8. The basic unit of heredity is the

A ribosome
B cell
C enzyme
D gene

9. Tallness (T) is dominant over shortness (t). What is the ratio of tall to short offspring from a cross between Tt and Tt?

A 1 : 1
B 9 : 3 : 3 : 1
C 3 : 1
D 1 : 2 : 1

10. Which blood vessel carries blood away from the heart?

A Artery
B Venule
C Vein
D Capillary

11. The functional unit of the kidney is the

A villus
B alveolus
C nephron
D neuron

12. Which enzyme in saliva begins the digestion of starch?

A Pepsin
B Salivary amylase
C Lipase
D Trypsin

13. Bats are classified as mammals because they

A are active at night
B can fly
C eat insects
D have fur and feed their young with milk

14. The part of a flower that develops into the fruit is the

A sepal
B anther
C ovary
D petal

15. The vector of the malaria parasite is the

A housefly
B tsetse fly
C female Anopheles mosquito
D male Culex mosquito

Answers and explanations

Mark your paper first, then read the explanation for every question you missed.

1. **D**	2. **C**	3. **C**	4. **C**	5. **D**
6. **A**	7. **B**	8. **D**	9. **C**	10. **A**
11. **C**	12. **B**	13. **D**	14. **C**	15. **C**

1. D. Aerobic respiration takes place in the mitochondria.

2. C. Only plant cells have a cellulose cell wall.

3. C. Chlorophyll absorbs light for photosynthesis.

4. C. Plants release oxygen as a by-product.

5. D. Yeast, not muscle, produces ethanol.

6. A. Green plants are producers.

7. B. The lizard eats the primary consumer.

8. D. Genes are sections of DNA that carry hereditary information.

9. C. TT, Tt, Tt and tt: three tall to one short.

10. A. Arteries carry blood away from the heart.

11. C. Each kidney has about a million nephrons.

12. B. Salivary amylase breaks starch into maltose.

13. D. Fur and milk are features of mammals; flying is not.

14. C. After fertilisation the ovary becomes the fruit.

15. C. Only the female Anopheles mosquito transmits the malaria parasite.